

BOLSHENARYMSKOE, KazakhstanWMO: **364280**Lat: **49.200N**Lon: **84.517E**Elev: **401**StdP: **96.60**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.3	-33.4	-38.9	0.1	-36.2	-36.1	0.1	-33.3	8.2	-6.3	6.3	-7.6	1.0	30	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	32.3	19.0	30.5	18.6	28.9	18.0	20.7	28.7	19.8	27.6	19.0	26.7	2.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.6	23.9	17.1	12.8	23.2	16.2	12.1	22.5	62.0	28.7	58.7	27.6	55.9	26.7	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	6.4	5.1	DB	-37.8	35.6	3.5	1.8	-40.3	36.9	-42.4	37.9	-44.4	38.9	-47.0	40.3	(4)
(5)			WB	-38.1	23.0	3.6	2.3	-40.6	24.6	-42.7	25.9	-44.7	27.2	-47.3	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-20.6	-17.7	-7.7	6.0	13.4	19.3	21.1	19.3	12.9	5.5	-4.5	-14.6	(6)	
	DBStd	15.57	7.54	6.81	7.53	5.44	4.22	3.55	2.92	3.39	4.34	4.61	6.75	7.07	(7)	
	HDD10.0	3791	948	774	548	138	17	0	0	0	19	149	436	762	(8)	
	HDD18.3	5891	1206	1008	807	369	160	32	9	29	168	397	686	1021	(9)	
	CDD10.0	1168	0	0	0	18	122	278	344	288	107	11	0	0	(10)	
	CDD18.3	226	0	0	0	0	7	60	94	58	7	0	0	0	(11)	
	CDH23.3	2758	0	0	0	4	129	669	1002	791	159	3	0	0	(12)	
	CDH26.7	925	0	0	0	0	27	211	362	286	39	1	0	0	(13)	
(14) Wind	WSAvg	1.9	1.1	1.2	1.5	2.2	2.6	2.3	2.1	2.2	2.1	2.0	1.7	1.3	(14)	
(15) Precipitation	PrecAvg	410	20	15	20	32	47	47	54	42	37	38	33	26	(15)	
	PrecMax	605	46	31	44	86	141	107	184	124	119	94	71	55	(16)	
	PrecMin	251	4	2	8	4	6	9	18	13	2	9	4	11	(17)	
	PrecStd	78	11	8	9	19	30	22	33	25	27	19	18	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	1.9	11.8	23.7	29.8	33.1	34.6	34.9	30.6	21.5	10.9	3.4	(19)	
		MCWB	-2.4	0.2	5.9	13.5	17.4	18.9	20.5	19.1	16.0	11.7	5.9	0.9	(20)	
	2%	DB	-4.2	-2.6	7.1	20.7	26.6	30.9	32.2	32.1	27.3	18.6	8.3	0.0	(21)	
		MCWB	-4.9	-3.8	3.6	11.5	15.4	18.7	19.6	18.6	15.2	10.4	4.8	-1.2	(22)	
	5%	DB	-6.3	-5.0	4.2	18.2	24.4	29.1	30.4	29.9	24.9	16.0	6.0	-2.7	(23)	
		MCWB	-7.0	-6.0	2.1	10.2	14.3	18.4	19.3	17.9	14.3	9.3	3.2	-3.6	(24)	
	10%	DB	-8.7	-7.1	2.3	15.5	22.1	27.3	28.6	27.8	22.4	13.5	3.8	-4.5	(25)	
		MCWB	-9.3	-8.0	0.7	8.9	13.4	17.7	18.6	17.3	13.4	8.2	1.8	-5.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.3	0.1	6.5	14.3	18.2	21.5	22.7	20.9	17.7	12.6	7.1	1.2	(27)	
		MCDB	-1.7	1.7	10.6	22.1	27.7	29.3	30.5	30.4	28.3	19.4	9.2	2.5	(28)	
	2%	WB	-4.9	-3.7	4.0	12.2	16.6	20.2	21.2	19.6	16.1	11.0	5.2	-1.3	(29)	
		MCDB	-4.3	-2.6	6.6	19.0	24.4	28.3	29.0	28.8	25.3	17.6	8.1	-0.1	(30)	
	5%	WB	-7.0	-5.9	2.2	10.7	15.3	19.2	20.2	18.8	15.0	9.8	3.4	-3.6	(31)	
		MCDB	-6.3	-5.1	4.0	16.9	22.5	26.9	27.5	27.8	23.7	15.1	5.8	-2.7	(32)	
	10%	WB	-9.3	-8.0	0.9	9.3	14.2	18.3	19.4	17.9	13.7	8.5	1.9	-5.3	(33)	
		MCDB	-8.7	-7.1	2.2	14.8	20.7	25.4	26.5	26.2	21.3	13.0	3.7	-4.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.8	11.6	12.4	13.3	12.8	12.7	14.4	14.5	11.5	9.1	9.2	(35)	
		MCDBR	9.1	11.6	11.2	17.6	16.8	15.7	15.7	17.9	18.4	16.3	11.2	9.3	(36)	
	5% WB	MCWBR	8.6	10.5	8.4	9.6	8.1	6.5	6.2	7.3	9.0	9.4	7.8	8.1	(37)	
		MCWBR	9.2	11.6	10.6	15.6	14.4	13.7	13.6	15.7	17.0	14.4	10.4	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.282	0.314	0.363	0.411	0.398	0.400	0.413	0.384	0.346	0.342	0.313	0.285	(40)	
		taud	2.116	2.063	2.038	2.107	2.225	2.299	2.279	2.332	2.394	2.296	2.165	2.082	(41)	
	Ebn at Noon		747	809	830	831	861	860	842	851	852	778	698	674	(42)	
		Edh at Noon	83	113	138	145	134	126	126	114	97	89	79	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.65	2.78	4.18	5.07	5.99	6.51	6.08	5.61	4.20	2.41	1.43	1.22	(44)	
	RadStd		0.11	0.10	0.34	0.44	0.47	0.51	0.44	0.34	0.40	0.25	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page